# Capstone Project: Employee Management System

## Project Overview

Develop an Employee Management System using Core Java (J2SE) with an emphasis on object-oriented programming principles such as abstraction, polymorphism, inheritance, and encapsulation. The system will include validation and exception handling for inputs, looping, use of collection interfaces, and abstract classes. The project will also involve basic input, processing, saving results into a MySQL database, retrieving, and displaying data. Additionally, it will include unit testing and a web interface using HTML, JavaScript, CSS, and Spring Boot.

**Functional Requirements**

1. **User Authentication**
    - Login, Logout
2. **Employee Management**
    - View, Add, Update, Delete, Search employee
3. **Input Parameters**
    - Employee ID
    - Name
    - Date of birth
    - Department
    - Salary
4. **Processing**
    - Calculate the average salary of employees
    - Calculate the average age of employees
5. **Database Operations**
    - Save employee details into a MySQL database
    - Retrieve employee details
    - Display employee details
6. **Report**
    - Display list of employees
        1. By Department
        2. By Age

**Technology Stack**

- **Frontend**: HTML, JavaScript, CSS
- **Backend**: Core Java (J2SE), Spring Boot
- **Database**: MySQL
- **Testing**: JUnit

**Implementation Details**

1. **Object-Oriented Programming Principles**
    - **Abstraction**: Create abstract classes for common functionalities.
    - **Polymorphism**: Use method overriding and interfaces.
    - **Inheritance**: Create a base class for common attributes and methods.
    - **Encapsulation**: Use private fields and public getter/setter methods.
2. **Validation and Exception Handling**
    - Validate user inputs (e.g., non-empty fields, valid age range).
    - Handle exceptions (e.g., invalid input, database connection errors).
3. **Collection Interface**
    - Use collections (e.g., ArrayList) to manage employee data.
4. **Abstract Class**
    - Create an abstract class Person with common attributes (e.g., name, date of birth).